

Preferential Accretion onto Eccentric and Unequal Binary Black Holes Systems

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ABSTRACT

Supermassive Binary Black Holes (SMBBHs) are expected to be surrounded by Circumbinary Disks (CBDs) which affect the binary through gravitational torques and accretion. It has been reported that the binary can experience “preferential accretion” where one black hole (BH) out-accretes the other for hundreds of orbits, but the phenomenon has yet to be fully described or understood. In this work, we utilize a suite of 80 SMBBH hydrodynamic simulations with varying mass ratios (q_b) and eccentricity (e_b) in order to robustly delineate the behavior of preferential accretion, determine its relationship to the structure of the CBD, and study its observational consequences. In this paper we report the most extensive series of measurements for q , the rate of change of the binary mass ratio, $\dot{q}_b$, as a function of e_b and q_b . We also show that a) the accretion behavior onto one BH over the other is strongly tied to the precession of the CBD, b) there can be sub- and super-Eddington accretion in a single binary, and c) there exists a regime where the CBD can drive the binary away from $q_b = 1$. Given these findings, we suggest flickering jets and an under-density of LISA events $q_b = 1$ as potential observable consequences.

1 INTRODUCTION

Cosmic structure forms hierarchically and galaxy mergers are expected to result in gravitationally bound Super-Massive Binary Black Holes (SMBBHs) (White & Rees 1978; Begelman et al. 1980). During galactic merger, it is also expected that the inter-stellar medium from the proto-galaxies are funneled to the galactic center. This creates a reservoir of material that, due to the conservation of angular momentum and the binary’s gravitational potential, forms a circumbinary disk (CBD). Many aspects of the CBD have been well studied in the literature, in both the SMBBH and star-planet contexts. It has been shown that the binary can influence the CBD, creating eccentric structure and precessing eigenmodes (Lubow 1991; Whitehurst 1994; Nelson 2003; Goodchild & Ogilvie 2006; Paardekooper et al. 2008; Kley et al. 2008; Shi et al. 2012; Miranda et al. 2017a; Thun et al. 2017; Muñoz & Lithwick 2020; Lubow 2022; Siwek et al. 2023a) and that the CBD can in turn influence the binary—changing both its semi-major axis and eccentricity (Tiede et al. 2020; Zrake et al. 2021; D’Orazio et al. 2013; Siwek et al. 2023b; D’Orazio & Duffell 2021; Farris et al. 2014), as well as exciting precession of the binary itself (Tiede et al. 2024; Dittmann et al. 2023; Calcino et al. 2023). However, crucially, the binary also accretes from the circumbinary disk, influencing both the mass-ratio and associated light-curves of the system (e.g. Siwek et al. 2023a; Westernacher-Schneider et al. 2022).

[MSS: I would suggest adding an “e.g., ...” to the above lists of citations (and some of the following) to convey that they are not exhaustive.] [SOD: done] Also Farris+2014, Duffell+2019

It has been a widely known result of CBD studies that the accretion onto the binary ($\dot{M}_b = \dot{M}_1 + \dot{M}_2$) is not steady but experiences variability through time. Near equal-mass ratio ($q_b \gtrsim 0.7$) circular binaries display a sawtooth pattern with a period of about 5 binary orbital periods (τ_b) potentially due to the presence of an $m = 1$ over-density traveling at the inner edge of the CBD (MacFadyen

& Milosavljević 2008; Miranda et al. 2017b; Muñoz et al. 2019; Westernacher-Schneider et al. 2022). Eccentric and unequal mass-ratio binaries too display variability on the order of a binary period (Miranda et al. 2017b; Muñoz et al. 2019; Westernacher-Schneider et al. 2022). [MSS: But only for equal mass, circular binaries! In eccentric binaries or circular binaries with mass ratios less than 0.7 or so, the variability is again dominated by terms on the order of the binary period, and the ‘lump’ disappears.] [SOD: done.] In order to study how the binary, as a function of q_b , accreted material from the circumbinary disk, Tiede et al. (2022) tracked the paths of passive tracer particles in a two-dimensional grid-based hydrodynamical code. They determined that much of the gas is viscously transported from the outer disk before adopting a ballistic trajectory and ultimately accreting on to the BHs near the inner edge. They also delineate an “accretion horizon” of 1.01 the semi-major axis of the binary (a_b), a radius that past which any material enters is accreted.

However, studies have also pointed out the discrepancy between the accretion behaviors of the primary and secondary components of the SMBBH (e.g. Miranda et al. 2017b; Siwek et al. 2023a; D’Orazio et al. 2024). Namely, the secondary tends to accrete, on average, at a greater rate than the primary in what has become known as “preferential accretion”—pushing the system to equal mass ratio ($q_b = 1$) (Miranda et al. 2017b; Siwek et al. 2023a). While systems with $q_b = 1$ generally accrete at equal rates, suggesting a stable equilibrium at $q_b = 1$, studies have reported that at certain eccentricities (e.g., $e_b = 0.5$ and $e_b = 0.6$), the binary can undergo transient “symmetry breaking.” (see Figure 7 in Muñoz et al. 2019). In these instances, one black hole temporarily accretes more than its counterpart before its accretion rate is suppressed, while the other experiences an enhanced accretion rate episode. This process was explored in Siwek et al. (2023a) by studying the accretion rate across 50 2D hydrodynamic simulations of SMBBHs in CBDs, sampling different binary parameters q_b and e_b .

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individual
found that can
 Siwek et al. (2023a) reported time averaged values of $\lambda = \dot{M}_2/\dot{M}_1$, the ratio of the secondary's accretion rate to the primary's, for their simulation suite and displayed snapshots of the corresponding CBD. Siwek et al. (2023a) report that low-eccentricity binaries display larger values of λ than high-eccentricity binaries ($e_b > 0.6$). Further, the process of the flip in preferential accretion was suggested to be associated with the behavior of the CBD. Namely, it was associated with a unique disk behavior they called "forced precession". DeLaurentiis et al. (2025) also discussed the preferential accretion rate of the binary, suggesting it be understood in geometric terms through the distance from the BHs to the CBD. *through*

The protoplanetary disk community have run two and three dimensional smoothed particle hydrodynamic (SPH) simulations of CBDs to *near the second ary* understand the accretion onto the primary and secondary (Günther & Kley 2002; Ochi et al. 2005; Young et al. 2015). In simulations run for $\leq 100\tau_b$ they found that the primary can out-accrete the secondary due to streamlines that, despite entering the Lagrange point L2, have such high angular momentum that they are able to flow to the *define* counterpart (Ochi et al. 2005; Young et al. 2015). Similar findings were suggested in Tiede et al. (2022). *to date*

Despite these efforts we are still yet to fully characterize and understand the rich detail of accretion onto the individual components of a binary in a CBD system. Thus, *add 2-2 sentence* to the following work we build on Siwek et al. (2023a) and *with the consent of the authors* use the larger simulation suite of Siwek et al. (2023b) to report the largest binary parameter sweep study on "preferential accretion". Further, we *characterize* the accretion behavior by applying unique numerical techniques and *some might* ~~use the existing analytic tools~~.

In Section 2 we briefly discuss the technical details of the simulations, the key concepts from linear theory we draw upon, and the key techniques we employ. In Section 3 we present our findings: detailing the temporal behavior of preferential accretion and mass-ratio evolution and describe how that *these* varies with q_b and e_b . In Section 4 we detail the consequences our findings have on observations. We discuss how our results could suggest a jets that periodically switch on and off, "flickering", and a population of SMBBHs with $q_b \neq 1$. In Section 5 we summarize our key findings and discuss next steps.

2 ANALYTIC TOOLS AND NUMERICAL METHODS

In the following section we briefly describe the setup of the Siwek et al. (2023a,b) simulations, the concepts from linear disk theory we employed, and the numerical tools we leveraged.

2.1 Simulation Setup

In the following we briefly describe the setup for the simulations of interest, but please see Siwek et al. (2023a,b) for a more thorough discussion.

Siwek et al. (2023a,b) performed 2D hydrodynamical simulations of binary black holes embedded in a finite, locally isothermal disk with an α -viscosity of $\alpha = 0.1$ and an aspect ratio of $h = 0.1$. The disk is initialized according to a power-law function that places the inner edge of the cavity at $2a_b$ and the outer edge at $\approx 50a_b$. The binary-disk system spans a computational grid of $300a_b \times 300a_b$ with open boundary conditions, allowing the disk to viscously spread over time and reach a quasi-steady state. *density and temperature profiles*

The binary is modeled as two sink particles with a mass-ratio q_b and radii $r_s = 0.03a_b$, moving on a fixed Keplerian orbit of eccentricity e_b . The fraction of gas accreted from each gas cell at

each time step by the sink particle is $\gamma_0 \left(1 - \frac{r_{ij}}{r_s}\right)^2$ where r_{ij} is the radial distance from the j th sink particle to the i th gas cell. In addition to ~~the mass of the gas cell~~ the sink also accretes ~~the gas~~ linear momentum *from the gas cells*.

The simulations, conducted with the moving-mesh code AREPO (REF), used Voronoi tessellation to generate a grid of cells, and explored a wide parameter space for q_b and e_b across the sets $q_b \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ and $e_b \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. The simulations were run for 10,000 orbits, with surface density snapshots recorded at apocenter every 10 orbits. Accretion rates, however, were recorded independently at a much higher cadence, with a time-step of ≈ 0.01 orbits.

[SOD: ask from magda what determined this cadence..] [MSS: Good question. I chose this cadence to capture as much high frequency variability as possible, while avoiding any numerical artifacts on the order of the sound crossing time in the sink region. But I don't think there's any need to comment on this here.] [SOD: sounds good.]

2.2 Disk Theory

In order to illuminate the relationship between the accretion of the binary and the behavior of the CBD we must first characterize the time-variable behavior and attributes of the disk. Namely, we focus on the eccentricity and precession of the disk. *maybe delete? this can only operate in test particles (or do you mean shocks in gas stream crossings?)*

Gas eccentricity within CBDs is expected to grow through mechanisms such as eccentric Lindblad resonances (ELRs) or spiral shock pumping at the cavity edge (Lubow 1991; Whitehurst 1994; Paardekooper et al. 2008; Kley et al. 2008; Shi et al. 2012). While ELRs and spiral shocks promote eccentricity growth (Lubow 1991; Shi et al. 2012), viscous damping (Goodchild & Ogilvie 2006) and orbit crossings (Statler 2001) act to suppress it. Previous 2D simulations have found steady-state eccentricity profiles, indicating that these competing effects can reach equilibrium (Miranda et al. 2017a; Siwek et al. 2023a). Further, 2D and 3D hydrodynamical simulations have found significant eccentricity near the inner edge of the disk (the cavity), with eccentricity declining outward (MacFadyen & Milosavljević 2008; Miranda et al. 2017a; Siwek et al. 2023a; Ragusa et al. 2024). Similar trends have been seen in magneto-hydrodynamic simulations (Shi et al. 2012), suggesting that eccentricity is a robust characteristic of CBDs. *the inner regions of eccentric*

Additionally, simulations have demonstrated that the CBDs can precess (Nelson 2003; Shi et al. 2012; Miranda et al. 2017a; Thun et al. 2017; Siwek et al. 2023a), with precession frequencies often attributed to the eigenmodes of a Schrödinger-like equation for eccentricity evolution (Goodchild & Ogilvie 2006; Shi et al. 2012; Teyssandier & Ogilvie 2016; Lee et al. 2019; Muñoz & Lithwick 2020; Lubow 2022).

A robust study of the eccentricity and precession of both the bulk of the disk and the cavity in hydrodynamic simulations was conducted by DeLaurentiis & Rafikov (2025). Utilizing the same suite of simulations as this paper, they delineate the shape of the non-linear inner edge of the circumbinary disk, the cavity, by extracting the dominant Fourier modes of the associated isodensity contour.

[SOD: Will dig a bit more for papers that explicitly state this assumption of geometry...but am dubious. Also do you have access to this artymowicz paper? I could only find the ADS link and Rafikov's discussion of it.] [MSS: Unfortunately I also couldn't find it, sorry] While alternative explanations have been proposed (Artymowicz 1983), it has widely been assumed that the preferential accretion of the binary is related to the relative closeness of the

$$q_b \equiv M_2/M_1 \leq 1$$

is this the one cited in Zrake+2023?

BH

to be clear, this is apsidal (not nodal) precession

components to the CBD's inner edge (Rafikov 2016). In order to test this dependence we utilize the results for the cavity shape and its precession period from DeLaurentiis & Rafikov (2025) in order to build a time-dependent geometric picture of the binary in the cavity.

2.3 Numerical Techniques

As discussed earlier, in order to highlight the link between accretion and the behavior of the CBD, we are interested in understanding both as time-dependent quantities. The shape of the CBD is inherently time-dependent due to its precession¹. Since our simulation runs output snapshots every 10 orbits, the time-series associated with our CBD is constrained to a 10τ cadence.

The accretion rate is time-dependent. For each sink, the simulation tracks the mass accreted Δm at time-step t , with a maximum cadence of 0.01τ . As we are interested in comparing the accretion rate to the behavior of the CBD, we down-sample the high cadence accretion rate data to yield a time-series that matches the 10 orbit cadence of the CBD time-series. We do this by summing the instantaneous mass accreted over 10-orbit non-overlapping sliding windows and dividing it by the window size of 10 orbits. After we transform the accretion rates for each BH into the lower cadence of 10 orbits via the aforementioned procedure, we construct our preferential accretion quantity $\lambda(t) = \frac{\dot{M}_2(t)}{\dot{M}_1(t)}$, the ratio of the accretion rate of the secondary to the primary.

We also report the rate of change of the mass-ratio $\dot{q}(t)$. We define it as

$$\dot{q} = \frac{[1 + q_b(t)][\lambda(q_b) - q_b(t)]}{1 + \lambda(q_b)} \frac{\dot{M}_1 + \dot{M}_2}{M_1 + M_2} \quad (1)$$

where

$$q(t) = \frac{M_2(t)}{M_1(t)} \quad (2)$$

and the mass of each BH

$$M_j(t) = \sum_0^t (M_j(t=0) + \Delta m(t)) \quad (3)$$

is simply the running sum. Again, we note that all time-variable inputs are first transformed to be of 10-orbit cadence via the aforementioned procedure.

3 RESULTS

In the following sections we report our key results. Namely, in Section 3.1 we detail the behavior of $\lambda(t)$ and report its mean value and, if time variable, its period. In Section 3.1.1, we make some preliminary investigation into the CBD's effect on the accretion behavior. In Section 3.2 we calculate the rate of change of the mass ratio, and highlight an instance of the binary accreting away from unity.

3.1 Preferential Accretion

In Fig. 1 we plot our $\lambda(t)$ time-series for the entirety of our simulation suite. The figure is structured such that each panel represents a unique

¹ While the eccentricity and semi-major axis are strictly time-dependent, we find that for most systems the shape of the cavity achieve a steady state and its time-dependence is due to its apsidal precession alone.

1.0	S	V	S	V	V	V	V	V
0.9	S	V	S	S	V	V	V	V
0.8	S	V	S	S	V	V	V	V
0.7	S	V	S	S	V	V	V	V
0.6	S	V	S	S	V	S	V	V
0.5	S	V	S	S	S	S	V	V
0.4	S	V	S	S	S	S	S	V
0.3	S	V	S	S	S	S	S	V
0.2	S	V	S	S	S	S	S	S
0.1	S	S	S	S	S	S	S	S
q_b/e_b	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.8

Table 1. Grid showing whether $\lambda(t)$ for a given binary is time-stable (S, Red) or time-varying (V, Blue).

The most striking feature simulation. The rows represent simulations of the same mass-ratio q_b , whereas the columns represent simulations of the same eccentricity e_b . The left-most columns are the least eccentric binaries, and the upper-most rows are the most equal-mass binaries.

What first strikes us in Fig. 1 is the broad division of $\lambda(t)$ into being constant or time-varying. Broadly, many low e_b , low q_b binaries are time-stable. Though some experience a sharp change in behavior in the first 2000 orbits associated with an expected initial instability (Moriwaki & Nakagawa 2004; DeLaurentiis & Rafikov 2025), they soon reach a near-constant value. This time-stable behavior is exemplified by the $e_b = 0.3$, $q_b = 0.3$ binary². Others show time-variable behavior with oscillations in $\lambda(t)$ spanning two orders of magnitude, such as $e_b = 0.6$, $q_b = 1.0$.

[SOD: is this too wishy washy and not quantitative enough?] In Table 1 we delineate whether $\lambda(t)$ is time-varying or time-stable via a blue cell with a V or red cell with an S, respectively. It is of particular note how similar Table 1, which depicts the time-variability of $\lambda(t)$, is to Table 1 of DeLaurentiis & Rafikov (2025) which depicts the precession of the CBD. [MSS: Can you clarify which table you're referring to here? I don't see this in the paper you cited.] [SOD: this paper was meant to be sent to you also, apologies] It seems as though the simulations with time-stable $\lambda(t)$ values map directly to the simulations that have a precessing CBD—suggesting that the preferential accretion of the binary is, in some part, determined by the behavior of the CBD and thereby the cavity. Siwek et al. (2023a) described the disk as exhibiting three distinct regimes: free precession, forced precession, or a locked disk. Through symmetry arguments, Siwek et al. (2023a) suggested that freely precessing or locked disks correspond to a time-averaged λ of unity, while disks undergoing forced precession show preferential accretion.

The $e_b = 0$ simulations break this one-to-one mapping. Though $e_b = 0$ simulations have been well documented (Miranda et al. 2017b; Siwek et al. 2023a) to have precessing disks, it is quite clear that their preferential accretion is time-stable. Due to the one-to-one mapping for non-circular binaries, this suggests that a binary may need to have an apocenter in order for the CBD behavior to influence its accretion. This is somewhat similar to the argument made in DeLaurentiis et al. (2025) which suggested that there needed to be a

² Though some binaries (e.g. $e_b = 0$, $q_b = 0.3$) experience noticeable modulation around this constant value, the dichotomy between time-varying and time-stable $\lambda(t)$ values remain clear.

Since x and y-axis range and labels are the same in each panel, you don't need to include these in each panel. Instead, you can save a lot of space by making the panels touch. S. DeLaurentiis et al. Also label must be at least the same size as the main text.

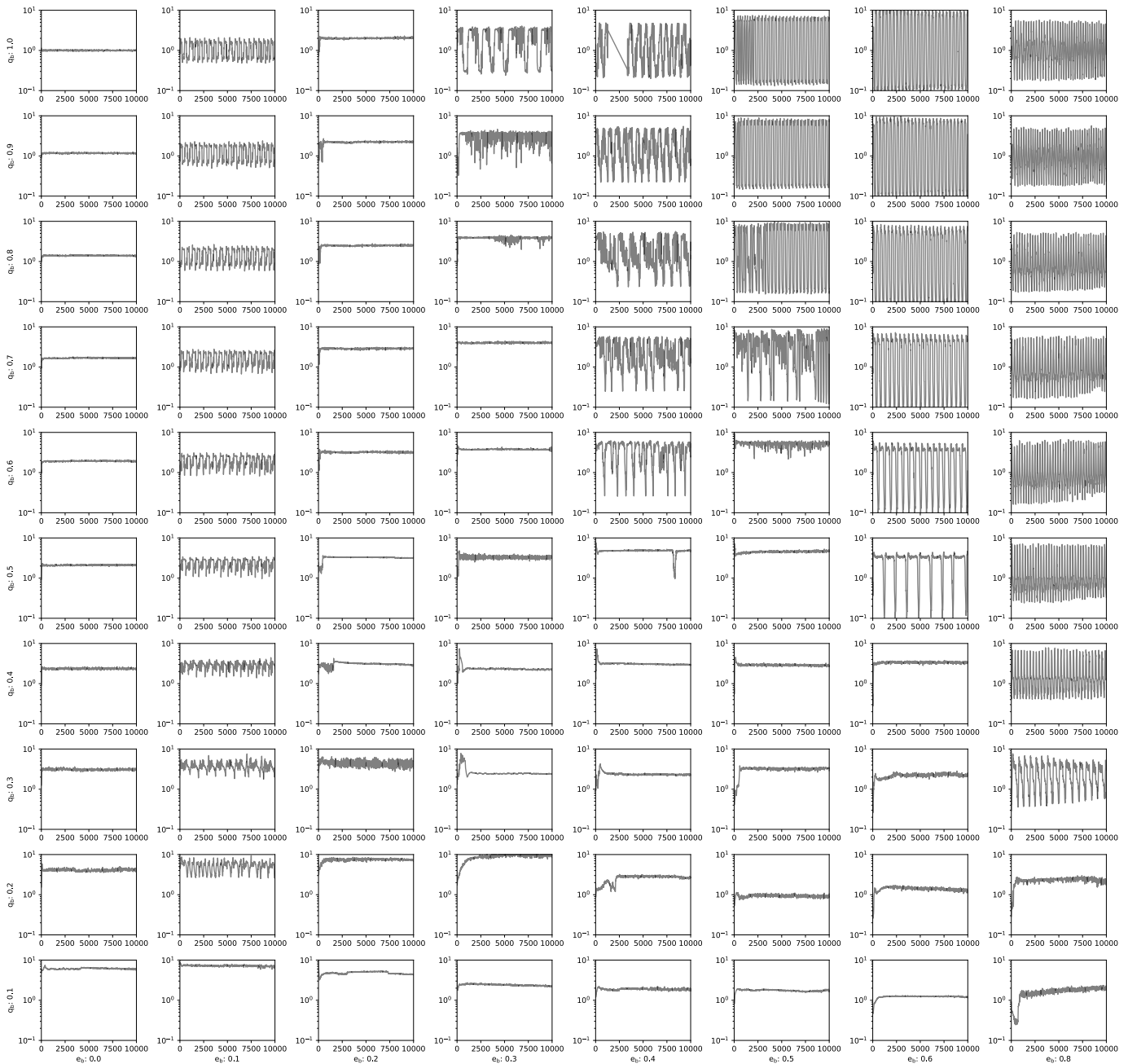


Figure 1. $\lambda(t)$ for our binary simulation suite. Each panel represents a unique simulation, with $\lambda(t)$ on the y-axis and time on the x-axis. The panel's position on the larger grid signifies its binary parameters, with e_b being constant along columns and increasing from left to right, and q_b being constant along rows and increasing from bottom to top.

minimum eccentricity in precessing binary simulations to see strong modulations in $\lambda(t)$.

[MSS: I also discussed the symmetry breaking in quite some detail in section 3.5 in Siwek+23a, including the relationship between disk precession period and the 'preferential accretion switching' – might be worth mentioning here.] [SOD: I included some more context, but from my reading of your paper you seem to suggest that locked disks have $\lambda = 1$, and $e_b = 0$ have $\lambda = 1$. I think this study shows that this is not the case. please correct me if i'm wrong, and let me know how it is best to approach this.]

"is a clear correlation"

Further, we note that the amplitude of $\lambda(t)$ oscillations are not uniform among time-varying simulations. In fact, we note that there seems to be a relationship between the amplitude of the $\lambda(t)$ oscillation and the e_b of the binary. [MSS: Nice!]

In Fig. 2 we display a heat-map indicating the variability of $\lambda(t)$. The cell values are colored according to the standard deviation of $\lambda(t)$. We note that while many of our simulations have small

³ For our calculations of the behavior of $\lambda(t)$ we make a cut at 3000 orbits, to not let the initial time-instability of the disk (DeLaurentiis & Rahkov 2025) influence our metrics.

can you explain this? do you mean orbit #5 3000 to 3000+N were used? N=?
you need to spend -2-3 sentences to explain the setup: these sims are different because binary has pre-imposed (GR) precession ---

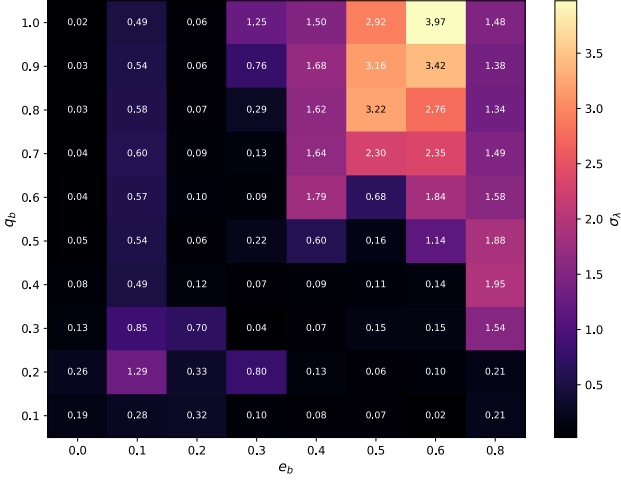


Figure 2. A heat-map displaying σ_λ , the standard deviation of $\lambda(t)$, to indicate the difference in oscillation behavior across our simulations.

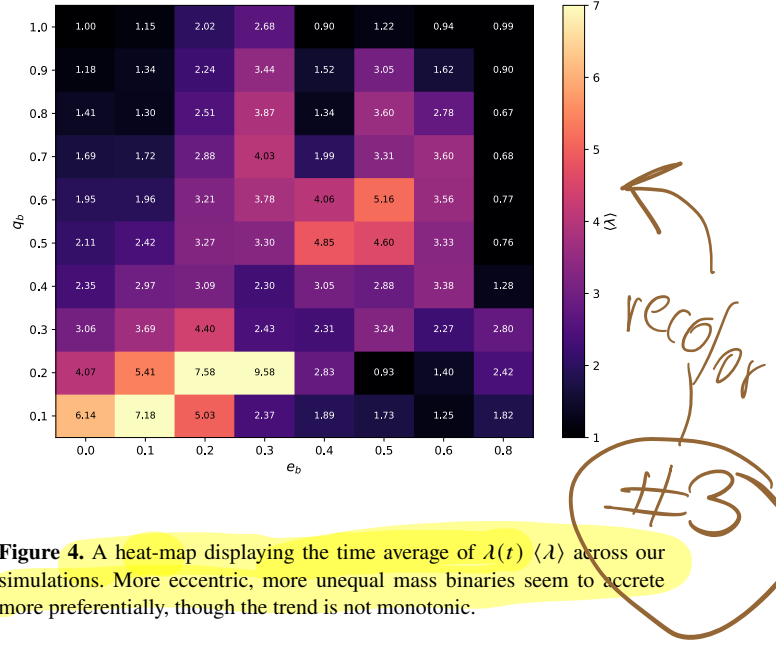


Figure 4. A heat-map displaying the time average of $\lambda(t)$ $\langle\lambda\rangle$ across our simulations. More eccentric, more unequal mass binaries seem to accrete more preferentially, though the trend is not monotonic.

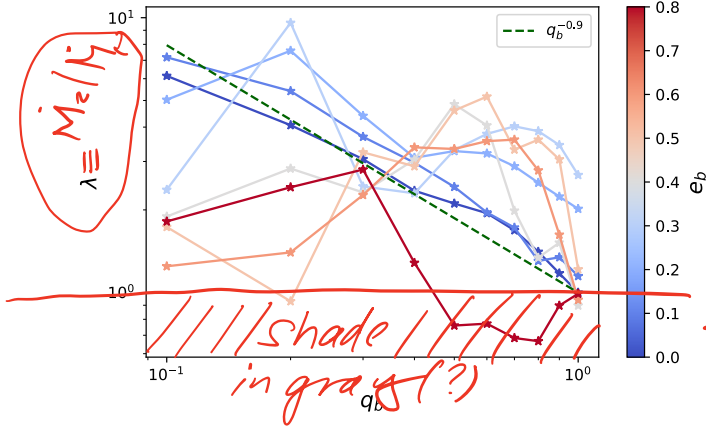


Figure 3. $\langle\lambda\rangle$ across q_b and e_b . $\langle\lambda\rangle$ is represented on the y-axis, q_b is on the x-axis, and the color represents e_b . We also plot the power-law fit to $\langle\lambda\rangle$ of $q_b^{-0.9}$ from Siwek et al. (2023a) in the dashed green line.

σ_λ since they are time-stable (see Table 1), those that display meaningful σ_λ suggest a trend. Namely, we find that σ_λ increase with e_b , peaking at $e_b = 0.6$. This modest increase in σ_λ with e_b is reminiscent of Figure 18 of DeLaurentiis & Rafikov (2025) which indicates that the eccentricity of the cavity increases with e_b and peaks at $e_b = 0.6$. This correlation further suggests that the CBD is a crucial regulator of preferential accretion.

In addition to temporal nature of $\lambda(t)$ we also comment on the mean value of $\lambda(t)$, determining which BH is preferred to accrete and to what extent. We report our results in Fig. 3 and Fig. 4.

In Fig. 3 we display the mean values of $\lambda(t)$ on the y-axis and the q_b on the x-axis, while the color represents e_b . [MSS: I like the idea of using a colorbar to differentiate eccentricities! However, I would suggest a sequential colorbar that would make it easier to distinguish the increasing eccentricities from one another. E.g. the ‘cool’ colormap in sequential (2) <https://matplotlib.org/stable/users/explain/colors/colormaps.html>] [SOD: done]. We find that our figure is in broad agreement with Figure 4 of Siwek et al. (2023a).

In particular, we too find that the $\langle\lambda\rangle$ of the $e_b = 0$ simulation seems to follow the power law of $q_b^{-0.9}$. However, we also find that $e_b = 0.1$ also holds true to this line, suggesting that the behavior of $\langle\lambda\rangle$ is generic to lower eccentricity binaries and not unique to $e_b = 0$. [SOD: should we actually do a regression and report error values on this?] Further, all our unequal-mass simulations suggest that if there is preferential accretion it is always to the secondary BH. This is in line with the results of the literature (Muñoz et al. 2019; Duffell et al. 2020; Farris et al. 2014; Dittmann & Ryan 2021; Siwek et al. 2023a). Despite preferential accretion being biased towards the secondary, we note that the extent to which one BH accretes over the other depends greatly on the binary parameters. The behavior of $\langle\lambda\rangle$ is perhaps best shown in Fig. 4.

Figure 4 displays q_b on the y-axis and e_b on the x-axis, plotting the value of $\langle\lambda\rangle$ according to color. On first glance, it is clear that some binary parameters lend themselves to stronger preferential accretion rates than others. Namely, in Fig. 4 we find that low e_b and low q_b values display high preferential accretion values, while high e_b and high q_b display lower ones as discussed in Siwek et al. (2023a). This is the opposite of the trend we saw for σ_λ values in Fig. 2 and for the eccentricity of the CBD (DeLaurentiis & Rafikov 2025). This suggests that the binary parameters themselves are more important than the CBD in regulating the level of preferential accretion.

While there is certainly a trend, the level of preferential accretion is by no means monotonic with respect to q_b and e_b . We note that intermediate values (e.g. $e_b, q_b = (0.4, 0.4), (0.5, 0.5)$) have higher preferential accretion levels than their neighbors. Further, Fig. 4 displays interesting ‘hotspots’ of high preferential accretion, with $e_b = 0.3, q_b = 0.2$ being the simulation with the highest levels of preferential accretion. We also find curious dimspots where the accretion onto both BHs is near-equal despite the binary having unequal mass. Notably, $q_b = 0.2, e_b = 0.5$ has a $\langle\lambda\rangle = 0.93$.

More importantly, we note a point of deviation from Siwek et al. (2023a). In Fig. 3 and Fig. 4 we report $\langle\lambda\rangle \neq 1$ for $e_b, q_b = (0.2, 1.0), (0.3, 1.0)$ simulations. The $e_b = 0.2, q_b = 1.0$ simulation in Siwek et al. (2023a) had a numerical error, resulting in a deviation in λ . Accounting for this, we determine that for $e_b, q_b = (0.2, 1.0), (0.3, 1.0)$ simulations, the binary, in fact, preferentially



Figure 5. The ratio of the period of preferential accretion (τ_λ) to the period of the CBD precession, τ_{r_1} . The x-axis is the eccentricity of the binary, the y-axis is the mass-ratio of the binary, and the color of the cells correspond to the ratio of the periods. The uncolored cells are those where a period for r_1 or λ could not be determined.

accretes onto one of the BHs. The implications of this are further discussed in Section 3.2 and Section 4.

Clearly, there is a variability in both the time behavior of preferential accretion and its value across simulations. In the following section we make a preliminary attempt to understand preferential accretion as an effect of the unique geometry between the binary and the CBD.

3.1.1 Preliminary Analysis

In Section 3.1 we noted similarities between the preferential accretion behavior $\lambda(t)$ and the CBD geometry. The delineation of time-stable and time-varying $\lambda(t)$ in Table 1 was likened to the precessing and locked circumbinary disks. σ_λ seemed to display the same trend with e_b and q_b as that of the CBD eccentricity. While alternative explanations have been proposed (Artymowicz 1983), the assumption that the preferential accretion of the binary is related to the relative closeness of the components to the CBD's inner edge (Rafikov 2016; Farris et al. 2014) has been left unchallenged. [MSS: Farris+14 also discuss this in their paper] [SOD: added.] In the following section we provide a first, preliminary, analysis of this assumption and alternative ways in which the CBD would influence preference accretion. [MSS: Nice!]

As discussed in Section 2, we use the dominant Fourier modes of the cavity to reconstruct its shape as a function of θ : $r_{\text{cav}}(\theta)$. Combined with the position of the BHs, we then define the distance

this is not defined. Where is "r" measured from? The center of mass? or from each BH?

from the inner edge of the CBD, "the cavity wall", to each BH as a function of θ . We take θ_1^* and θ_2^* such that $r_{\text{cav}}(\theta)$ is minimized for each BH. We then calculate $r_1 = r_{\text{cav}}(\theta_1^*) - r_{\text{BH}_1}$ and $r_2 = r_{\text{cav}}(\theta_2^*) - r_{\text{BH}_2}$ where r_1 and r_2 are the closest distances from the cavity-wall to the primary and secondary BH, respectively. Since our BH positions are fixed—as we only record snapshots at apocenter— r_1 and r_2 simply encode the information about the size and orientation of the CBD and serve as proxies.

A first step in determining the relationship between preferential accretion and the CBD is to study the temporal behavior. We have already determined that, except for the $e_b = 0$ case, all precessing CBDs result in a non-time stable $\lambda(t)$. $r_1(t)$ and $r_2(t)$, proxies for the CBD orientation, display the same split into time-stable and time-varying behavior as $\lambda(t)$ in Table 1. For simulations with time-varying $\lambda(t)$ we compare their period of oscillation with that of r_1 and r_2 . To determine the period of each time-series we first calculate a fast Fourier transform (FFT) and normalize the amplitudes by their sum. The period is defined at that of the largest amplitude in the spectrum which has a normalized value > 0.05 . We find the periods of r_1 ⁴ and λ in this fashion and display the ratio of the two periods in Fig. 5.

We note that the ratios reported in Fig. 5 are nearly all equal to one. Though certain binaries have ratios that deviate from unity we note that upon further investigation these deviations do not suggest resonances between CBD precession and preferential accretion but rather point to the difficulty of FFT to handle periodic, non-sinusoidal $\lambda(t)$ for such simulations (e.g. $e_b = 0.4$, $q_b = 0.8$). Thus, we find that the precession of the CBD determines not only whether $\lambda(t)$ varies but also the period at which it varies.

In Fig. 6 we display r_1 (blue) r_2 (red) and λ (black) for a slice in time for six of our simulations, with the dual y-axis representing the dimensionless lambda value (left) and the distance from each BH to the edge of the cavity in units of a (right).

As indicated by Fig. 5 the variability of the distance from the BHs to the cavity displays the same period (or lack thereof) as λ . This is especially clear for the simulations in the left column of Fig. 6. However, we note that it is also true for some of the simulations in the right column where the period is time-varying. It is especially well captured by $e_b = 0.5$, $q_b = 0.7$ where the change from a stochastic to periodic λ signal at $\approx 8500\tau$ is mirrored in the change of behavior of r_1 , and r_2 . [SOD: can we say that its stochastic there? how do we better phrase this?] Further, in $e_b = 0.4$, $q_b = 0.7$ we find that large amplitude changes in r_1 and r_2 (e.g. $\approx 6000, 9000\tau$) occur simultaneously with large amplitude changes in λ .

The naive understanding of the tie between preferential accretion and the distance to the black holes is as follows: if one black hole is significantly closer to the inner edge of the circumbinary disk it should, by way of its greater tidal potential on the gas, be able to pull more gas towards it faster and thereby have a greater accretion rate. In $e_b = 0.2$, $q_b = 0.1$ we see what we would expect from our naive explanation. The panel [MSS: is this still figure 6?] shows that this simulation has a CBD that is locked such that our primary black hole (blue) is about twice as far from the cavity edge than our secondary black hole (red). Correspondingly, we see that the secondary is accreting at a rate about five times the primary, seemingly as a result of the secondary's greater pull on its neighboring portion of the CBD. In fact, we can even take this further and note that the tidal potential on the gas by the secondary is about four to

⁴ It is important to note that we have found the period of r_1 and r_2 to be exactly equal in all our simulations, making either appropriate for this calculation.

make panels bigger by concatenating them

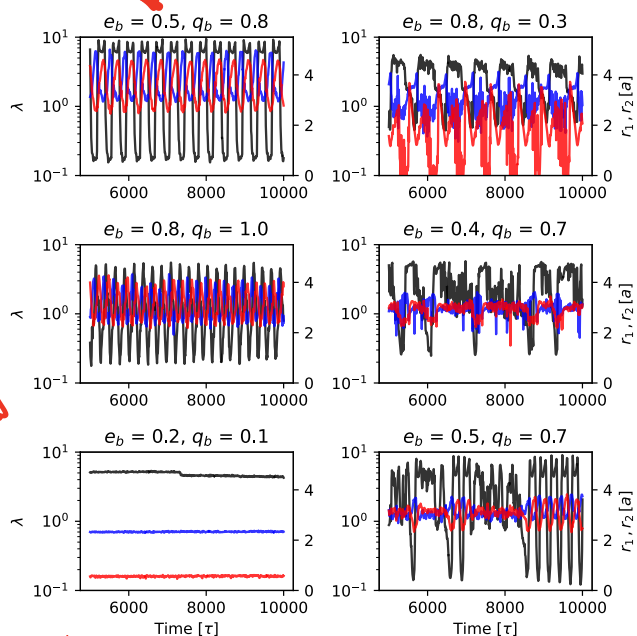


Figure 6. Comparison between the time-series of $\lambda(t)$ (black), $r_1(t)$ (blue), and $r_2(t)$ (red). Clockwise from upper left we display $e_b, q_b = (0.5, 0.8)$, $(0.8, 0.3)$, $(0.4, 0.7)$, $(0.5, 0.7)$, $(0.2, 0.1)$, and $0.8, 1.0$. We find that the BHs position with respect to the cavity does not determine the components' relative accretion rates.

five times greater than its companion, the same factor difference as the accretion rate. However, our naive approximation is inherently over-simplistic. It doesn't take into account each BH's minidisk, the streams by which the gas is fed to the BH, or any non-linear fluid dynamics, such as shocks, that may be relevant in this mechanism. Studying the other panels, we see that our naive explanation no longer suffices.

All simulations displayed in Fig. 6, except for $e_b = 0.2, q_b = 0.1$, show that lower r_2 values do not correspond to higher accretion rates onto the secondary. The naive explanation suggests that r_1 should be exactly "in-phase" with λ while r_2 should be exactly $\pi/2$ "out-of-phase". However, we find that $(e_b, q_b) = (0.5, 0.8)$ and $(e_b, q_b) = (0.8, 1.0)$ both clearly do not display such behavior. The former displays that both r_1 and r_2 are $\pi/4$ out of phase with λ , while the latter displays the exact opposite to our naive expectation, that r_1 is in fact "in-phase" and r_2 is "out-of-phase". Further, when turning to $e_b, q_b = (0.8, 0.3)$ and $e_b, q_b = (0.4, 0.7)$ we note that our naive approximation further breaks down, as now the amplitude of r_1 relative to r_2 does not strictly coincide with one BH having preferred accretion over the other. We note that for $(e_b, q_b) = (0.8, 0.3)$ while the primary is always at least as far away from the cavity-wall as the secondary, we still see periods where the primary accretes preferentially. Further, for $(e_b, q_b) = (0.4, 0.7)$ we see a similar issue in that during periods where both the primary and secondary are equidistant from the cavity wall, the secondary is accreting up to 4 times more than the primary⁵.

From the above analysis it is clear that the naive explanation

"A striking feature"

⁵ While λ is a ratio and this behavior could be due to small amplitude deviations in $\dot{M}_1$ and $\dot{M}_2$, we found that the amplitude of the sum is not strongly suppressed during this period.

why "opposite"? this is in fact what you state two sentences earlier as the expectation (although I didn't understand the reason...)

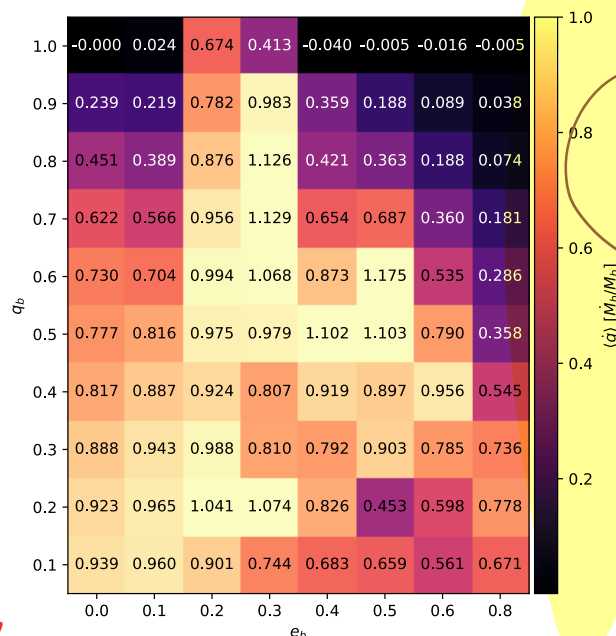


Figure 7. The average rate of change of the mass-ratio $\langle \dot{q} \rangle$ for our simulation suite. The x-axis is e_b , the y-axis is q_b , and the value of $\langle \dot{q} \rangle$ is represented as color.

of distance to the cavity and tidal potential is not robust. While the variability and periods of both are linked, they are far from causal. [SOD: is there a way to make this punchier?] Preliminary further analysis has suggested that the deviations from the behavior can be understood through the lens of ballistic trajectories and shock induced accretion. While this study has focused on describing the zoo of preferential accretion behaviors and its ties to the CBD, a more robust explanation on how the binary preferentially accretes will be explored in a forthcoming paper.

3.2 Mass Ratio

In addition to preferential accretion, we also report the average rate of change of the mass ratio, $\langle \dot{q} \rangle$. Namely, we calculate $\dot{q}(t)$ as described in eqn. xx in Section 2, make a time-cut at 3000τ to ensure that early instabilities do not effect our results, and report the mean on the truncated time-series in units of $\dot{M}_b/\dot{M}_b$ in Fig. 7.

What first strikes us in Fig. 7 is that for all $q_b \neq 1$ simulations the binary $\langle \dot{q} \rangle$ is positive, meaning that the binary is evolving towards equal mass. [MSS: I would suggest a diverging colormap here (e.g., 'seismic') to distinguish between growing and decreasing mass ratios.] [SOD: they are all positive so i don't see how we could colorcode..] We note, however, that the rate of this evolution is quite varied, spanning two orders of magnitude. Further, the gradient of $\langle \dot{q} \rangle$ in parameter space is not uniform. In fact, we see that there exists a massive drop off in $\langle \dot{q} \rangle$ in the upper right corner of Fig. 7, starting along the diagonal extending from $q_b = 0.8, e_b = 0.6$.

Turning to the upper row of Fig. 7 we find that while most of the $q_b = 1$ simulations display near zero values of $\langle \dot{q} \rangle$ ⁶ we find that there are two simulations which notably do not. Namely, the

⁶ The negative values reported are within the standard error of the mean for the $\dot{q}$ time-series ($\approx 10^{-2}$ for $q_b = 1$) and can thus be taken to be zero.

how did you determine this?

#3

the two are correlated, there is no clear causal link between them.

#5

Eq. 1.

?? top row is negative (I agree w/ Magda)

(q_b, e_b) = (1, 0.2) and (1, 0.3)

equal-mass.

heh? what do you mean?

$q_b = 1, e_b = 0.2$ and $q_b = 1, e_b = 0.3$ simulations display strong positive $\langle \dot{q} \rangle$ values, meaning that the “secondary”⁷ BH is growing more massive than the primary. However, since this a $q_b = 1$ case, this non-zero $\langle \dot{q} \rangle$ means that the binary is in fact being fed in such a way that it is evolving away from unity. This result for $(e_b, q_b) = (0.2, 1.0)$ stands as a correction to Siwek et al. (2023a) where it was reported that q_b would remain at one for this case. However, after ensuring to correctly track the sink particles in the $q_b = 1$ simulations we find that accretion towards unity for SMBBHs is not necessarily a given. This makes sense especially in light of the results of DeLaurentiis & Rafikov (2025) where it was reported that the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations have stable, locked disks in line with Siwek et al. (2023a), with a stable non-varying λ . *that the* In this case, *this is* was coincident with the primary being further from the cavity edge than the secondary.

The finding that equal-mass binaries at $e_b = 0.2, 0.3$ accrete away from equal mass is an entirely novel result and has implications on CBD structure and SMBBH population statistics. Firstly, from DeLaurentiis & Rafikov (2025) we note both $(e_b, q_b) = (0.2, 1.0)$ and $(0.2, 0.9)$ simulations have locked disks in roughly the same orientation, with the pericenter of the disk being closest towards the secondary. However, we note that the $q_b = 1$ case is accreting away from unity in such a way that the BH which was initially the “secondary” becomes the primary. This would likely cause the disk, which is oriented towards the secondary, to realign itself—flipping according to the switch in primary and secondary position. The details of the disk’s reaction to this change in binary parameters would not only provide key insights into the mechanism behind CBD orientation, but could provide insight not only into how far away from unity the binary evolves, but depending on the process of the re-orientation could prove to be an event with characteristic EM signatures. *the* [MSS: can you give some estimates here on the timescale of disk re-alignment?][SOD: Perhaps the viscous time? I’m not sure what do you think, Zoltan?] *I would speculate this occurs on a typical disk precession timescale (since “flipping” will probably occur via precession)*

4 OBSERVATIONAL IMPLICATIONS

In the following section we discuss the observational consequences of our $\lambda(t)$ and $\dot{q}$ results (see Section 3). *some potential*

4.1 Flickering Jets

A key observational consequence of accretion onto BHs is the potential launching of jets. While the exact mechanism to launch jets continues to be an extremely active area of study, we can broadly understand jet launching through the lens of the radiative efficiency of the accretion disk. When an accretion disk is radiatively efficient the photons, and thus the energy they provide, are efficiently radiated away from the disk. However, in radiatively inefficient disk, the photons get trapped in the disk, causing a buildup of energy and pressure, both thermal and magnetic. This scenario can result in large pressure gradients that could drive jets, as well as a strong buildup in magnetic field strength which could be extracted via the Blandford-Znajek mechanism. Radiatively inefficient, geometrically thick disks are commonly found at low accretion rates $\dot{M} < 0.01 \dot{M}_{\text{Edd}}$ (Guolo

define $\dot{M}_{\text{Edd}}$

⁷ In the equal-mass case, “secondary” and “primary” merely identify the components of the binary which at apocenter are in the same spatial quadrant as the secondary and primary of an unequal mass binary.

et al. 2021) and at high accretion rates $\dot{M} > \dot{M}_{\text{Edd}}$. Given these criteria for jet-launching we are able to make statements about the jets from our simulations.

To do so, we must first scale our numerical accretion rates, which are in units $\dot{M}_{\text{bin}}/\tau$ to Eddington units. To be able to retain information about the relative accretion rates of each BH to the other, we must set the accretion rate of the binary $\dot{M}_b \equiv \dot{M}_1 + \dot{M}_2$ to be $\gamma \dot{M}_{\text{Edd}}$ where γ is an arbitrary scale factor and

$$\dot{M}_{\text{Edd}} \equiv \frac{4\pi G M m_p}{c \eta \sigma_T} \quad (4)$$

where m_p is the proton mass, σ_T is the Thomson scattering cross-section and $\eta \equiv 0.1$ is the radiative efficiency. We are then able to convert the accretion rate of each individual BH to units of $\dot{M}_{\text{Edd}}$.

In Fig. 8 [MSS: The axis labels here are unfortunately too small. You may need to take out some parameters to make it readable] we set the accretion rate of the binary to be $1 \dot{M}_{\text{Edd}}$ assuming a $10^7 M_\odot$ binary and plot the accretion rate for both BHs normalized to their respective Eddington accretion rates. The horizontal gray dashed line is at $1 \dot{M}_{\text{Edd}}$ to represent the accretion rate above which jets are likely to launch. The red lines represent the accretion rate of the secondary, and the black lines represent the accretion rates of the primary. The background color of the panel is associated with different jet-behaviors: purple for dual jets, blue for a single jet, green for flickering jets. The time-slice displayed is arbitrary and serves to merely highlight the accretion behavior.

What first strikes us in Fig. 8 is the wide variety of difference in magnitude between the two BHs’ accretion rates. Since the Eddington accretion rate scales linearly with the BH mass we expect that the accretion rate of the secondary to increase greatly when normalized to Eddington units. This is evidenced by the nearly 2 order of magnitude difference between the secondary and primary at $q_b = 0.1$. Further, we also notice that the behavior of the individual accretion rates of the black holes are quite varied, as the $\lambda(t)$ results suggested. Aside from the differences in whether the BHs’ accretion rates are stable or not, the profile of the accretion rate itself is varied. Some binaries experience accretion rates that are close to sinusoidal (e.g. $e_b = 0.6, 0.8$), others seem to closer resemble square-waves (e.g. $q_b = 1, e_b = 0.3, e_b = 0.5$), others yet have quite sharp breaks that evade simple characterizations (e.g. $e_b = 0.1, q_b = 0.7$). Further, we note that the accretion rate of one BH is not always simply the accretion rate of the other with a different baseline and $\pi/2$ phase shift. Rather, they can take on notably different profiles from each other, resulting, at times, in both BHs experiencing a local peak in accretion rate, but because of different accretion rate amplitudes result in a peak in λ . It is *already said above* this zoo of individual BH accretion rates, and the way in which they compare to each other, that yield an interesting assortment of jet-launching behaviors.

In Fig. 8 we delineate three broad regimes: a) binaries where one BH launches a jet (blue), b) binaries where both BHs coincidentally launch jets in a sustained and repeated fashion (purple), c) binaries where both BHs launch jets in a successive, alternating fashion (green). We describe these jet-behavior regimes as single jets, dual jets, and flickering jets, respectively. We assigned jet-regimes by determining whether the accretion rate of each BH surpassed a threshold value of $1.1 \dot{M}_{\text{Edd}}$ for greater than 50 τ , and whether those instances were temporally coincident for greater than 50 τ .

We note that the majority of simulations at the assumed binary accretion rate of $1 \dot{M}_{\text{Edd}}$ are single-jet systems, where most of them have low q_b due to the highly unequal Eddington-normalized accretion rates of those systems.

The flickering jet systems are clustered at higher q_b and e_b . Due

I believe key is that these disks become geometrically thick and the inner

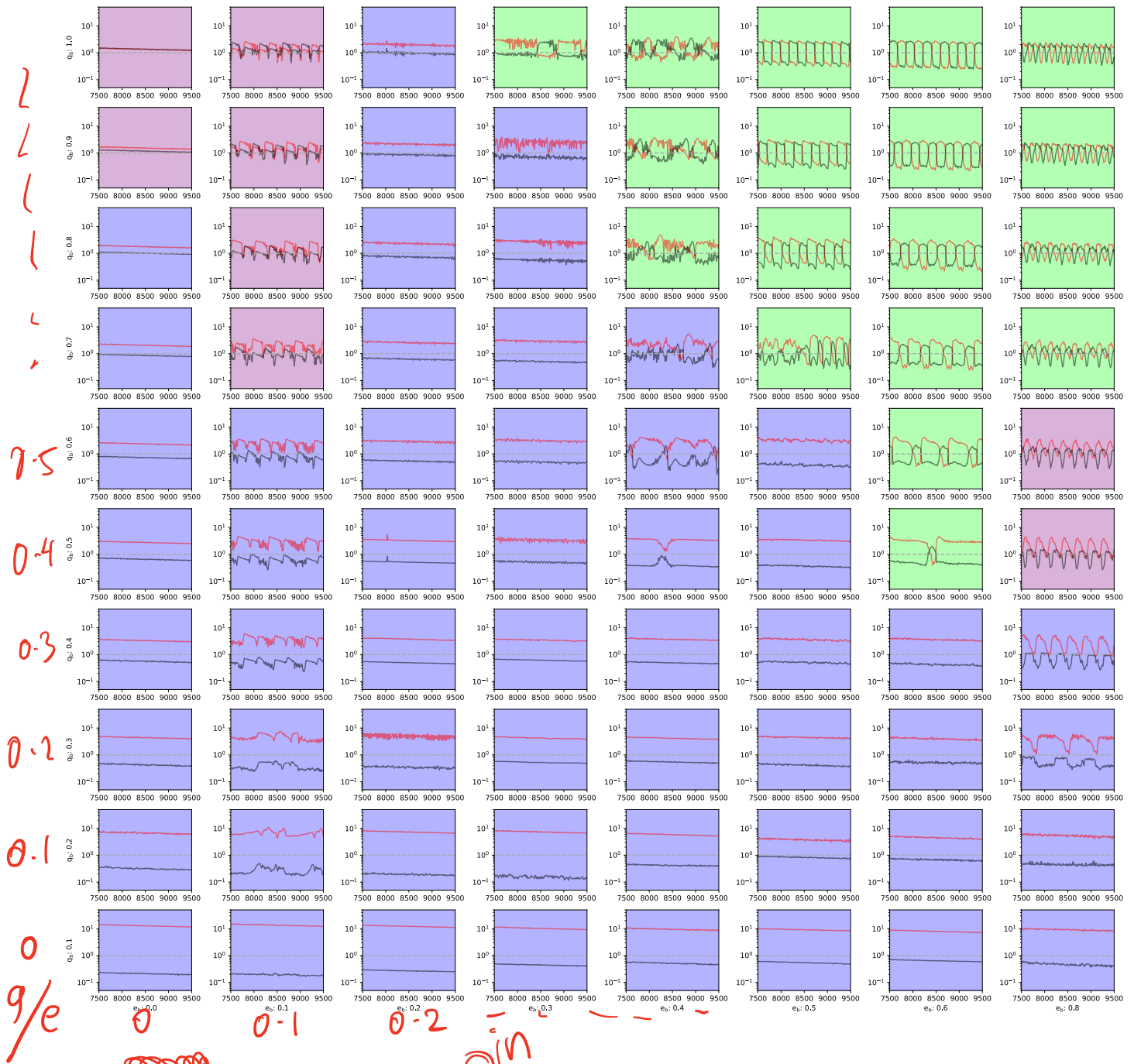


Figure 8. Eddington-normalized accretion rates for our simulation suite, with panel backgrounds colored by jet-regime. The panel structure is the same as that of Fig. 1, with time on the x-axis of each panel, accretion rate, in Eddington units, on the y-axis. We display the accretion rate of the primary (red), the secondary (black) and the jet-launching threshold of $1 M_{\text{Edd}}$ (horizontal dashed gray). Blue background colors indicate single jet regimes, purple indicates dual-jet regimes, and green indicates flickering jet regimes.

to large λ amplitudes, such systems are able to exist up to $q_b = 0.5$. It is important to note that since these flickering jets are dependent on large λ oscillations they are unique to $e_b \neq 0$ binaries.

Finally, we have a few dual-jet systems. For those at $e_b = 0$ both BHs accrete at nearly the same, stable rate and thus both launch sustained jets. However, we also note that we have dual jets for binaries whose $\lambda \neq 1$. We note that the dual-jet systems at $e_b = 0.1$ and $e_b = 0.8$ show both BHs having coincident, local peaks in accretion rates, meaning that the binary sustainably and periodically launches two jets.

While dual jets have been suggested before (Palenzuela et al. 2010;

Qian et al. 2019) and have even been simulated (Gutiérrez et al. 2024; Ressler et al. 2025), we believe that we are the first to report evidence suggesting a flickering-jet behavior. [MSS: But are the timescales of several hundred binary periods the same as those reported in the papers you cite? Or are they different? If so, why?] [SOD: no, previous discussions have not focused on the mechanisms of launching dual jets. they just assume initial conditions such that jets are launched. Further, the dual-jet case is assuming both are continuously launched simultaneously.] An entirely new regime, it not only provides a unique observational determination of eccentric

also Stu Shapiro has old MHD sims on binary jets. Also Elias Most...

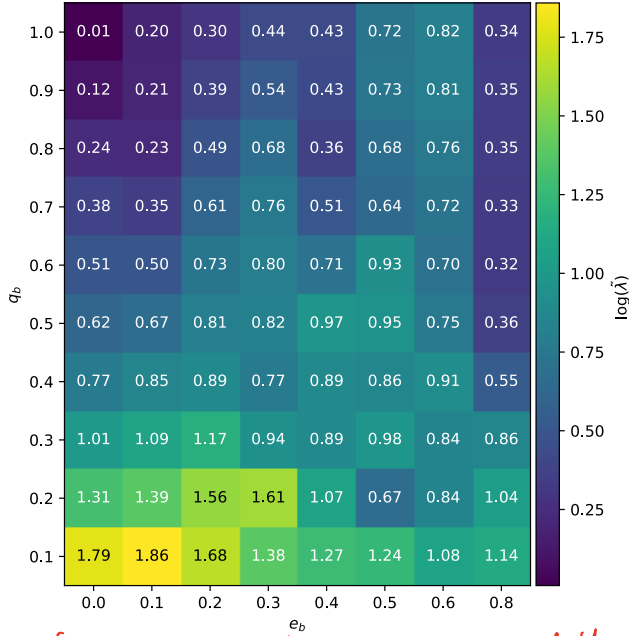


Figure 9. $\tilde{\lambda}$ for our simulation suite. The x-axis is e_b , the y-axis is q_b , and the colors indicate the value of $\log(\tilde{\lambda})$. ("Accretion rate range")

binaries but it is also a uniquely distinct kind of feedback mechanism that could be further explored in cosmological simulations.

Fig. 8 and its discussion assumed a binary accretion rate of $1\dot{M}_{\text{Edd}}$. However, such assumption controls the baseline of the binary accretion rate. While a different assumed binary accretion rate would not change the profiles nor the relative accretion rates of the components, it would affect the absolute amplitude—changing its jet-launching behavior.

Fig. 9 displays $\tilde{\lambda}$, which can be written as

$$\tilde{\lambda} \equiv \frac{\langle \max(\dot{M}_1, \dot{M}_2) \rangle}{\langle \min(\dot{M}_1, \dot{M}_2) \rangle} \quad (5)$$

where max and min are element-wise functions and the accretion rates are provided in Eddington units.

The result $\tilde{\lambda}$ is dimensionless and provides the ratio of the maximum and minimum Eddington-normalized accretion rates of the binary components. Given the time-variability of λ in Table 1 and a binary accretion rate, $\tilde{\lambda}$ can then be easily used to determine whether one or both BHs launch a jet, and whether they are temporally coincident. Namely, one can use

$$\dot{M}_1 = \frac{\dot{M}_b}{\tilde{\lambda} + 1} \quad (6)$$

to find the accretion rate of the primary based on the scaling of λ and $\dot{M}_b$. With Fig. 9 in hand we are able to gain insight into the jet behavior of any proposed BBH system as a function of q_b and e_b . Given the clustering of jet-behavior in q_b - e_b parameter space, determination of whether a system has sustained dual jets, flickering jets, or a single jet could greatly constrain the orbital parameters of the system.

“Flickering jets” are a unique EM signature for BBHs. In order to use them to find BBH systems, we must ensure they “flicker”—i.e. switch which BH is preferentially accreting—on a human time-scale. In the following we compute and place constraints on the time to “flicker.”

highlight/point to the detection regions...

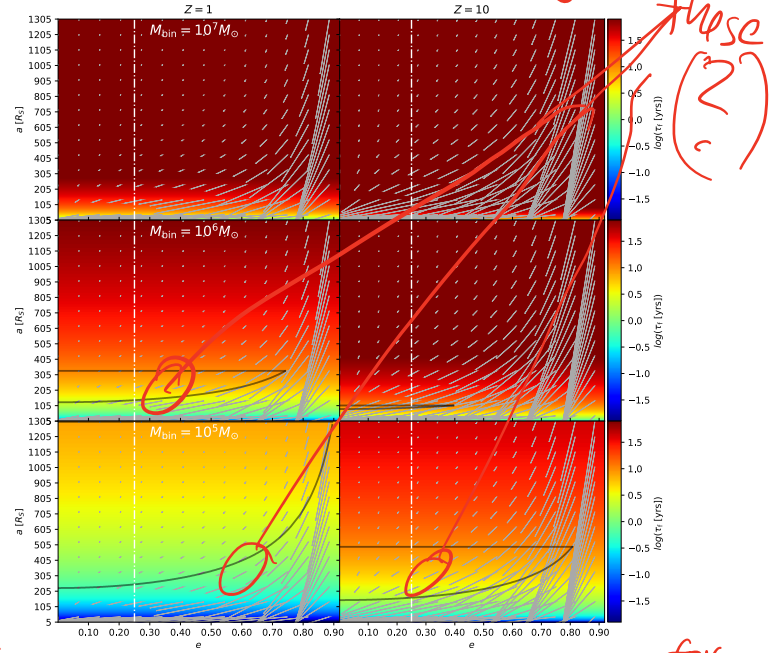


Figure 10. The region of parameter space that satisfies the time constraints (within black lines) for various binary masses at various redshifts. The x-axis of each panel is the eccentricity e_b , the y-axis is the a_b in Schwarzschild radii, and τ_f is reported in the observer frame. The gray lines represent the net effect of 10 orbits of GW radiation on the eccentricity and semi-major axis of each BBH.

Firstly, we require that the time to flicker τ_f be fewer than 10 years in the observer’s rest-frame, so that a few cycles could be possible in a human lifetime. For simplicity, we assume the flickering period is 300τ . We also require that the binary not merge in less than 100 years, in order to ensure that these systems are common enough.

In Fig. 10 we display the region of parameter space that satisfies the above time constraints (within black lines) for various binary masses at various redshifts. The x-axis of each panel is the eccentricity e_b , the y-axis is the a_b in Schwarzschild radii, and τ_f is reported in the observer frame. The gray lines represent the change in eccentricity and semi-major axis for the binary, due to 10 orbits worth of GW radiation.

Fig. 10 shows that $10^5 M_\odot$ and $10^6 M_\odot$ systems both display considerable regions of parameter-space that satisfy the constraints necessary to observe the flicker, up to $z = 10$. We find that the region which satisfies these constraints shrinks to zero by $z = 15$ and $z = 100$ for the $10^6 M_\odot$ and $10^5 M_\odot$ systems, respectively. This provides encouraging evidence for our ability to observe these flickering jet systems.

In addition to observing a flicker occur, we note that jets are extended emission sources and thereby provide us an ability to observe a past flicker. If we could determine a geometric separation in the structure of a helical jet, this could indicate that the emission is from a binary that flickered in the past. [MSS: I really like this figure! To make it more readable, can you increase the size of the axis labels and the text in the subplots indicating the binary mass?]

[SOD: should we talk about changing look spectra?...there is evidence that $z=0.1$ at 10^7 also has region, it is very small though]

4.2 Unequal Mass Sources

In addition to the variety of jet launching behavior and their effects on galaxy feedback, the evolution of the mass ratio of these SMBBHs is also a key observational signature. We explore whether our findings from Fig. 7 suggest that some binaries do not evolve towards equal mass in fact lead to detectable $q \neq 1$ BBH systems.

In addition to its mass-ratio, the eccentricity and semi-major axis of the binary also change as functions of q_b and e_b due to the CBD. We represent these as $\dot{e}_{\text{gas}}(q_b, e_b)$ and $\dot{a}_{\text{gas}}(q_b, e_b)$, respectively. The values for $\dot{a}_{\text{gas}}(q_b, e_b)$, and $\dot{e}_{\text{gas}}(q_b, e_b)$ were also calculated for this full simulation suite and are reported in Figure 2 and 3 of Siwek et al. (2023b), respectively. Since our simulation suite is discretized in e_b and q_b space, we define $\dot{e}_{\text{gas}}(q_b, e_b)$, $\dot{a}_{\text{gas}}(q_b, e_b)$ and $\dot{q}(q_b, e_b)$ as sets of piecewise functions with constant values that are centered at the binary parameter values and extend to the midpoint between it and its neighboring, sequential, binary parameter value. For example,

$$\dot{q} = 1.041 \begin{cases} 0.15 \leq q_b < 0.25 \\ 0.15 \leq e_b < 0.25 \end{cases} \quad (7)$$

However, in addition to CBD effects, the GWs radiated from the system also circularize and shrink the binary. We include these effects through the terms

$$\dot{a}_{\text{GW}} = \frac{-64G^3 M^3 q f(e)}{5c^5 a^3 (1+q)^2} \quad (8)$$

and

$$\dot{e}_{\text{GW}} = -e \frac{304G^3 M^3 q \left(1 + \frac{121}{304} e^2\right)}{15c^5 a^4 (1-e^2)^{5/2} (1+q)^2} \quad (9)$$

Thus, we are left with the set of coupled differential equations

$$\dot{a}(q_b, e_b) = \dot{a}_{\text{gas}}(q_b, e_b) + \dot{a}_{\text{GW}}(q_b, e_b, a_b) \quad (10)$$

$$\dot{e}(q_b, e_b) = \dot{e}_{\text{gas}}(q_b, e_b) + \dot{e}_{\text{GW}}(q_b, e_b, a_b) \quad (11)$$

$$\dot{q}(q_b, e_b) = \dot{q}_{\text{gas}}(q_b, e_b) \quad (12)$$

$$(13)$$

which we numerically integrate given an initial a_0 , q_0 , and e_0 .

Fig. 7, we solve the initial value problem for two equal-mass binaries with $e_b = 0.2$ and $e_b = 0.3$, integrating forward for $10^4 \tau$ with $a_0 = 10^3 R_S$. The results are displayed in Fig. 11.

In Fig. 11 we display the evolution of the two binary's $e(t)$ and $q(t)$ in the upper and lower panels, respectively. Firstly, our numerical integration confirms the expectation that the binary evolves towards the expected $e_b = 0.45$ equilibrium eccentricity (Duffell et al. 2020; Zrake et al. 2021; Siwek et al. 2023b, 2024). [MSS: You might also cite my 2024 paper on massive black hole binary population statistics, where we explored this in detail]. [SOD: done.] More interestingly, we find that the two binaries initialized at $e_b, q_b = (0.2, 1.0)$ and $(0.3, 1.0)$ do indeed evolve away from $q_b = 1$, and have end values of $(0.45, 0.991)$ and $(0.45, 0.998)$, respectively. These results suggest that BBHs, which at some point in their evolution, have the parameters $(0.2, 1.0)$ and $(0.3, 1.0)$ have equilibrium mass-ratios that are not equal to one. LISA is expected to have a precision in q_b of up to 0.5% [MSS: What accretion rate did you assume here? I would also consider changing the x-axis in figure 11 to semi-major axis, if that's feasible to do.] [SOD: i assumed 100*eddington, but the result shouldn't change correct? just how quickly it gets to the steady state. also, a doesn't change strongly over this time.] [SOD: Try as i might, i was unable to

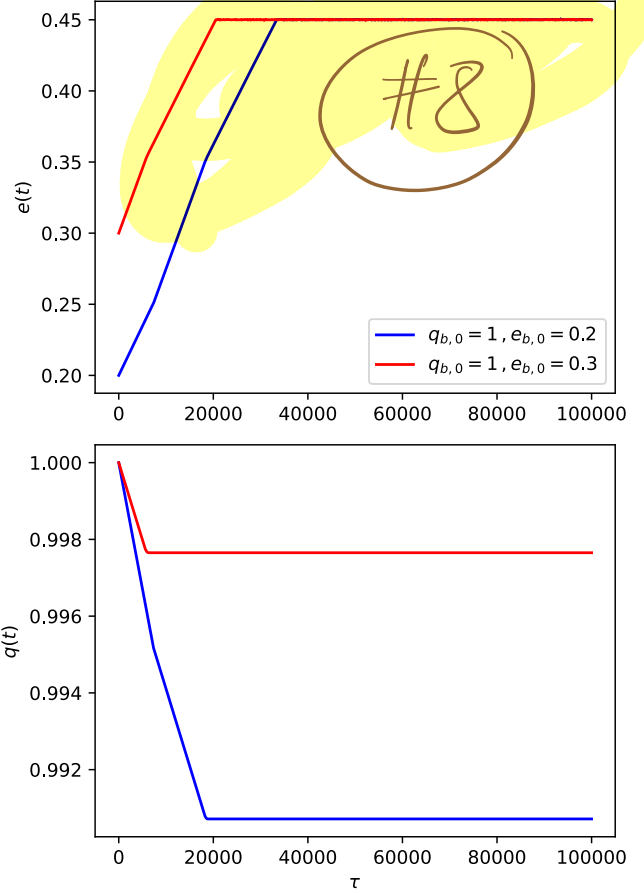


Figure 11. Evolution of binaries away from equal-mass. We depict the result of our numerical integration of equation 10 for $e_{b0}, q_{b0} = (0.2, 1.0)$ (blue) and $(0.3, 1.0)$ (red). The y-axes are $e(t)$ (upper) and $q(t)$ (lower) of the binary and the x-axes are time in units of each binary's orbital period.

find/remember the source for the LISA q precision.] Thus, while the deviation from $q_b = 1$ for the end state of $(0.3, 1.0)$ binaries may not be detectable by LISA, that of $(0.2, 1.0)$ will be within detection limits. Thus, given enough LISA detections, a number density bump at $q < 1$ could suggest the history of the binaries, indicating it passed through $(q_b = 1, e_b = 0.2)$.

5 SUMMARY AND CONCLUSIONS

This paper has provided the most extensive report on preferential accretion $\lambda \equiv \frac{\dot{M}_2}{\dot{M}_1}$ and mass ratio rate of change $\dot{q}$ for SMBBHs embedded in CBDs. In addition to calculating quantities, we provide novel insight into the behavior of these values over time and their dependence on q_b and e_b . We also conduct a preliminary investigation into how the CBD regulates preferential accretion. We summarize our key findings below.

- Across q_b and e_b , $\lambda(t)$ can be split into time-varying and constant regimes, broadly mirroring the split of the CBD into precessing and locked states. (Table 1)
- The average preferential accretion value $\langle \lambda \rangle$ exhibits a weak positive correlation with both e_b and q_b . The standard deviation of λ , σ_λ , exhibits a weak negative correlation with both e_b and q_b . (Fig. X)

- (iii) $\langle \lambda \rangle$ is positively correlated with more eccentric CBDs (Fig. 8)
- (iv) We find strong evidence that while the CBD precession and $\lambda(t)$ oscillation have the same period, these time-series are not in phase.
- (v) We do not find evidence that a BH must be closer to the CBD cavity than its counterpart, to accrete preferentially at a higher rate
- (vi) Normalized to Eddington accretion rates, $\lambda(t)$ results in disparate accretion regimes for each BH in the binary—leading to unique jet-launching regimes. We delineate binaries that are likely to launch a sustained jet from one BH (single-jet), from each BH (dual-jet), or alternate in which BH launches a jet (flickering-jet).
- (vii) We find that $e_b, q_b = (0.2, 1.0)$ and $(0.3, 1.0)$ binaries do not have $q_b = 1.0$ steady-states and preferentially accrete away from equal-mass. The steady-state of a $(0.2, 1.0)$ system will be detectable by LISA as having unequal mass.

While our work has shed light on novel aspects of this system, we are ~~not~~ studying setups with simplified physics, namely two-dimensional hydrodynamic simulations. In future work we hope to expand this study to more physically motivated simulation setups, accounting for effects that were ignored in this study. Namely, we hope to detail λ and $\dot{q}$ for radiation MHD simulations in three dimensions. Further, we would like to also develop GRMHD simulations for SMBBHs in order to better simulate jet-launching and test our findings on different jet regimes. In addition to developing more robust simulations, we would also like to run live-binary simulations for the $e_b, q_b = (0.2, 1.0)$ and $(0.3, 1.0)$ binary systems to not only better extrapolate how the binary evolves away from unity but to also understand how the disk reacts to the change in position of the primary and secondary. Finally, we will further investigate the mechanism behind preferential accretion, with a forthcoming analysis on the hydrodynamics of the gas within the cavity.

We conclude by noting that while the mechanism behind preferential accretion is more complex than previously thought, it is still intimately tied to the CBD and can have profound observational consequences for SMBBH systems, such as jet-launching and LISA population studies. We encourage further study into the preferential accretion mechanism and its consequences.

DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

REFERENCES

- Artymowicz P., 1983, *Postepy Astronomii Krakow*, **31**, 19
- Begelman M. C., Blandford R. D., Rees M. J., 1980, *Nature*, **287**, 307
- Calcino J., Dempsey A. M., Dittmann A. J., Li H., 2023, *arXiv e-prints*, p. [arXiv:2311.13727](https://arxiv.org/abs/2311.13727)
- D’Orazio D. J., Duffell P. C., 2021, *Astrophysical Journal Letters*, **914**, L21
- D’Orazio D. J., Haiman Z., MacFadyen A., 2013, *Monthly Notices of the Royal Astronomical Society*, **436**, 2997
- D’Orazio D. J., Duffell P. C., Tiede C., 2024, *ApJ*, **977**, 244
- DeLaurentiis S., Rafikov R., 2025, *arXiv e-prints*, p. [arXiv:2405.07897](https://arxiv.org/abs/2405.07897)
- DeLaurentiis S., Haiman Z., Westernacher-Schneider J. R., Krauth L. M., Davelaar J., Zrake J., MacFadyen A., 2025, *ApJ*, **980**, 55
- Dittmann A. J., Ryan G., 2021, *ApJ*, **921**, 71
- Dittmann A. J., Dempsey A. M., Li H., 2023, *arXiv e-prints*, p. [arXiv:2310.03832](https://arxiv.org/abs/2310.03832)
- Duffell P. C., D’Orazio D., Derdzinski A., Haiman Z., MacFadyen A., Rosen A. L., Zrake J., 2020, *ApJ*, **901**, 25

- Farris B. D., Duffell P., MacFadyen A. I., Haiman Z., 2014, *Astrophysical Journal*, **783**, 134
- Goodchild S., Ogilvie G., 2006, *Monthly Notices of the Royal Astronomical Society*, **368**, 1123–1131
- Günther R., Kley W., 2002, *A&A*, **387**, 550
- Guolo M., Ruschel-Dutra D., Grupe D., Peterson B. M., Storch-Bergmann T., Schimoia J., Nemmen R., Robinson A., 2021, *MNRAS*, **508**, 144
- Gutiérrez E. M., Combi L., Romero G. E., Campanelli M., 2024, *MNRAS*, **532**, 506
- Kley W., Papaloizou J. C. B., Ogilvie G. I., 2008, *Astronomy and Astrophysics*, **487**, 671
- Lee W.-K., Dempsey A. M., Lithwick Y., 2019, *The Astrophysical Journal Letters*, **882**, L11
- Lubow S. H., 1991, *Astrophysical Journal*, **381**, 259
- Lubow S. H., 2022, *Monthly Notices of the Royal Astronomical Society*, **516**, 5446–5453
- MacFadyen A. I., Milosavljević M., 2008, *ApJ*, **672**, 83
- Miranda R., Muñoz D. J., Lai D., 2017a, *Monthly Notices of the Royal Astronomical Society*, **466**, 1170
- Miranda R., Muñoz D. J., Lai D., 2017b, *MNRAS*, **466**, 1170
- Moriwaki K., Nakagawa Y., 2004, *Astrophysical Journal*, **609**, 1065
- Muñoz D. J., Lithwick Y., 2020, *Astrophysical Journal*, **905**, 106
- Muñoz D. J., Miranda R., Lai D., 2019, *ApJ*, **871**, 84
- Nelson R. P., 2003, *Monthly Notices of the Royal Astronomical Society*, **345**, 233
- Ochi Y., Sugimoto K., Hanawa T., 2005, *ApJ*, **623**, 922
- Paardekooper S. J., Thébault P., Mellema G., 2008, *Monthly Notices of the Royal Astronomical Society*, **386**, 973
- Palenzuela C., Lehner L., Liebling S. L., 2010, *Science*, **329**, 927
- Qian S. J., Britzen S., Krichbaum T. P., Witzel A., 2019, *A&A*, **621**, A11
- Rafikov R. R., 2016, *ApJ*, **827**, 111
- Ragusa E., Lynch E., Laibe G., Longarini C., Ceppi S., 2024, *Astronomy and Astrophysics*, **686**, A264
- Ressler S. M., Combi L., Ripperda B., Most E. R., 2025, *ApJ*, **979**, L24
- Shi J.-M., Krolik J. H., Lubow S. H., Hawley J. F., 2012, *The Astrophysical Journal*, **749**, 118
- Siwek M., Weinberger R., Muñoz D. J., Hernquist L., 2023a, *Monthly Notices of the Royal Astronomical Society*, **518**, 5059
- Siwek M., Weinberger R., Hernquist L., 2023b, *Monthly Notices of the Royal Astronomical Society*, **522**, 2707
- Siwek M., Kelley L. Z., Hernquist L., 2024, *MNRAS*, **534**, 2609
- Statler T. S., 2001, *The Astronomical Journal*, **122**, 2257–2266
- Teyssandier J., Ogilvie G. I., 2016, *Monthly Notices of the Royal Astronomical Society*, **458**, 3221–3247
- Thun D., Kley W., Picogna G., 2017, *Astronomy and Astrophysics*, **604**, A102
- Tiede C., Zrake J., MacFadyen A., Haiman Z., 2020, *Astrophysical Journal*, **900**, 43
- Tiede C., Zrake J., MacFadyen A., Haiman Z., 2022, *ApJ*, **932**, 24
- Tiede C., D’Orazio D. J., Zwick L., Duffell P. C., 2024, *Astrophysical Journal*, **964**, 46
- Westernacher-Schneider J. R., Zrake J., MacFadyen A., Haiman Z., 2022, *Phys. Rev. D*, **106**, 103010
- White S. D. M., Rees M. J., 1978, *MNRAS*, **183**, 341
- Whitehurst R., 1994, *Monthly Notices of the Royal Astronomical Society*, **266**, 35
- Young M. D., Baird J. T., Clarke C. J., 2015, *MNRAS*, **447**, 2907
- Zrake J., Tiede C., MacFadyen A., Haiman Z., 2021, *Astrophysical Journal Letters*, **909**, L13

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